Ethan Jeong

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EDUCATION

University of Maryland, College Park

B.S. in Computer Science | Expected Graduation: May 2028 | GPA: 3.57

- Dean's List (Fall 2024, Spring 2025)
- President's Scholarship
- Relevant Coursework: **Object-Oriented Programming, Computer Systems, Web Application Development, Algorithms, Data Science**

George Mason University

Dual Enrollment Program | GPA: 4.0 | Aug 2023 - Jun 2024

- Coursework: **Matrix Algebra, Multivariable Calculus**

TECHNICAL SKILLS & CERTIFICATIONS

- Programming Languages: Python(Pandas), Java, HTML/CSS, JavaScript, SQL, C, x86-64 Assembly, OCaml, Rust
- Frameworks & Tools: Git/GitHub, Flask, REST API(Twilio), Tkinter, and Globe.gl
- Certifications: CompTIA Security+ (SY0-701) | Aug 2026

EXPERIENCE

- **Student Researcher | FIRE Research | Aug 2024 – Dec 2025 | College Park, MD**
- Investigated how CO2 concentration affects methane production in Jug Bay wetland soil through two incubation trials with a three-person team.
- Prepared samples with different CO2 headspace concentrations (0 ppm, 400 ppm, 450 ppm, 500 ppm), measured methane at multiple time points over 10-day trial periods using gas chromatography, and analyzed results in Excel.
- Synthesized experimental findings and found that methane concentrations rose across all CO2 headspace treatments, reaching their highest final reading in the 500 ppm treatment; presented the findings as evidence of a possible relationship between added CO2 and methane production.

PROJECTS

Satellite Tracker | Python, HTML, CSS, JavaScript, Flask, Globe.gl | Jun 2026 – Sep 2026

- Built a full-stack web application with Flask, JavaScript, HTML, CSS, and Globe.gl that retrieves 17,000 active satellite Two-Line Element records from CelesTrak, providing a searchable dashboard with satellite details and an interactive 3D globe.
- Implemented orbital calculations to convert parsed TLE data from CelesTrak into latitude, longitude, and altitude using Earth-Centered Inertial (ECI) and Earth-Centered, Earth-Fixed (ECEF) coordinate systems.
- Developed a two-hour JSON caching and fallback system to reduce redundant CelesTrak requests and maintain application functionality when live satellite data was unavailable.

Daily Schedule SMS Messenger | Python, Twilio API, Tkinter, JSON | Feb 2025

- Built a Python application that uses the Twilio API to send automated SMS schedule reminders to users.
- Designed a modular system separating administrative capabilities, scheduling logic, and message creation and delivery to support easier maintenance and modifications.
- Built an admin-only Tkinter interface for managing users and creating, modifying, and deleting schedules.

Space Invaders Game | Java | Mar 2025

- Developed a 2D arcade game using object-orient-programming to build a custom polygon-based collision system that detects interactions between player projectiles(bullets), enemy projectiles, ships, and other game objects.
- Designed an object-oriented architecture using inheritance and interfaces to separate the player, enemies, projectiles, movement, rendering, and collision behavior into reusable components.
- Created a progressive difficulty system with randomized shooting, health, and scoring mechanics, and increasingly difficult rounds that adjust enemy speed and firing frequency.